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AraBERT-QC: a novel quantum-based classification architecture to classify short Arabic sentences

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Abstract

The use of quantum computing to analyze text is still at a nascent stage, especially when dealing with low-resource languages like Arabic, which has a notably intricate structure. The current state of quantum technology still limits the development of powerful models for processing Arabic text. This paper introduces AraBERT-QC, a novel hybrid model that integrates the Arabic language model AraBERT with a quantum classifier. We leverage AraBERT to generate contextualized sentence embeddings that capture the semantic structure of Arabic text, which are subsequently processed by the quantum classifier. Our experimental study is structured in two parts. The first part investigates the impact of various quantum data embedding techniques on the accuracy and efficiency of the quantum support vector machine (QSVM) algorithm. The second part explores the performance and efficiency trade-offs between two hybrid quantum-classical approaches: Hybrid quantum-classical neural networks (H-QNN) and the variational quantum classifier (VQC). We evaluate our architecture on two distinct datasets, sarcasm detection and sentiment analysis, using metrics such as accuracy, macro $F1$-score, precision, recall, and computation time. Results show that complex quantum embeddings achieve better performance in some tasks, while the VQC trains faster than the H-QNN. Notably, AraBERT-QC achieves 86.25% accuracy in sarcasm detection and 84.33% in sentiment analysis, outperforming the classical SVM and neural network baselines by up to 2% in accuracy and $F1$-score. This study highlights the potential of integrating quantum techniques with classical models to tackle challenges in computational linguistics, particularly for morphologically rich languages.

Keywords Quantum computing · Quantum machine learning · Quantum natural language processing · Arabic language processing · Text classification

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1 Introduction

Every day, millions of people generate data online at an unprecedented rate. This massive growth means we need smarter ways to process and understand all the information. In particular, text-based information has become essential for extracting insights from this digital explosion. As the internet becomes an essential part of everyday life, a tremendous amount of text is created across various platforms such as social media, blogs, news outlets, and customer reviews. Natural Language Processing (NLP) is an area of artificial intelligence that empowers computers to comprehend, interpret, and generate human language in a way that makes sense. This is achieved by combining computational linguistics with statistical methods, machine learning, and deep learning, enabling analysis of grammar, extraction of meaning, and generation of coherent language-based responses. It plays a key role in various fields like translating languages, understanding emotions in text, finding relevant information, and beyond, enabling efficient handling and processing of large-scale textual content [1, 2]. Recently, natural language processing has undergone significant transformation thanks to the development of large language models (LLMs). These models, built on sophisticated transformer-based architectures and trained on an extensive data, enabling them to perform a wide range of language-related tasks with unprecedented accuracy and efficiency. Their ability to generate coherent and contextually rich textual insights, coupled with their capacity to incorporate external knowledge, has revolutionized various domains, including content creation, customer support, and research applications [3]. However, despite their impressive capabilities, LLMs face notable limitations. One significant challenge associated with LLMs is their ability to effectively handle low-resource languages [4]. Arabic, in particular, poses unique challenges due to its rich morphology, diverse dialects, and complex syntactic rules. These intricacies are often underrepresented in the datasets used to train LLMs, which predominantly prioritize high-resource languages. As a result, the generated Arabic text is less precise and context-sensitive, which diminishes its practical usability in real-world situations. Another significant challenge lies in the extensive training requirements of LLMs. For instance, LLaMA 2, a state-of-the-art model developed by Meta, comprises an impressive 70 billion parameters. Training this model required about 71.7 days [5], highlighting the immense computational cost and making it difficult to fine-tune for low-resource languages like Arabic. The growing complexity of computational challenges is driving the need for more powerful systems than what traditional computers can offer. Quantum computers utilize the unique features of quantum mechanics to tackle problems that would take classical systems an impractically long time to solve. Achieving such performance showcases the potential of quantum technology to exceed classical limits, a concept known as quantum supremacy [6]. Quantum computing is an innovative method that takes advantage of quantum mechanics concepts, like superposition, entanglement, and interference, to handle and process information. In contrast to conventional computers that work with binary bits (either 0 or 1), quantum computers utilize qubits, which can simultaneously represent multiple states. This capability allows quantum computers to solve specific,

complex problems significantly faster than classical systems, particularly in optimization, artificial intelligence, and simulation tasks, making them highly specialized and transformative in fields like finance, chemistry, cybersecurity, and machine learning [7]. Quantum computing might revolutionize traditional machine learning by using novel quantum algorithms to overcome current challenges and boost performance in key areas. Techniques such as variational quantum circuits, quantum Monte Carlo integration, and quantum amplitude estimation have shown promise in improving optimization and clustering tasks. These methods offer advantages like better scalability in high-dimensional spaces and polynomial or quadratic speedups in convergence compared to classical approaches [8]. Furthermore, quantum-native approaches, such as quantum neural networks and quantum kernel methods, provide theoretical benefits like higher expressiveness for data modeling. However, practical implementation remains constrained by the limitations of current quantum hardware, which still lacks the robustness required for reliable, large-scale deployment. [9]. Despite notable progress, researchers continue to encounter significant challenges in quantum computation. The realization of fully fault-tolerant quantum computing remains a distant aspiration due to the immense resources required to implement effective error correction codes. Current noisy intermediate-scale quantum (NISQ) devices lack the robustness necessary for reliable, large-scale computations. At this stage, quantum systems are characterized by medium-sized processors built with inherently noisy qubits. This period bears resemblance to the early days of classical computing, where the interplay of analog and digital signals laid the groundwork for transformative advancements in information processing. Despite their inherent noise limitations, NISQ devices offer a sufficient number of qubits to enable experimental exploration of quantum computation's potential. These systems hold promise for achieving quantum advantages in domains such as quantum chemistry, variational quantum algorithms, quantum machine learning, and quantum optimization [10]. Additionally, developing effective quantum algorithms for near-term devices is inherently difficult. Many algorithms suffer from issues like the barren plateau problem, where optimization gradients diminish, impeding convergence [11, 12]. Quantum machine learning fuses the ideas of quantum computing with modern machine learning by tapping into quantum characteristics such as superposition and entanglement. This fusion enables innovative ways to handle and interpret data that can outperform traditional computational approaches. QML is currently being explored to optimize algorithms for solving complex problems more efficiently, including tasks like data classification, optimization, and dimensionality reduction. It holds immense potential across various domains, including quantum chemistry, material science, and financial modeling, where traditional approaches often face limitations [13]. Recent advancements in QML have also extended to natural language processing (NLP), particularly for widely spoken languages like English. Hybrid quantum-classical architectures and variational quantum models are being developed to tackle diverse challenges in NLP [14, 15]. However, research on low-resource languages, such as Arabic, remains limited.

In this work, we address key challenges in Arabic natural language processing by developing a hybrid model that merges classical deep learning with quantum computing techniques. We introduce a new architecture named AraBERT-QC, which

integrates the Arabic-specific transformer model AraBERT with a quantum-based classifier. This hybrid design aims to improve the performance of short Arabic text classification by leveraging the deep contextual understanding from classical models and the potential computational advantages offered by quantum systems. To evaluate the effectiveness of our model, we conduct experiments on two different datasets, targeting two core NLP tasks: sentiment analysis and sarcasm detection. Initially, we employ the pre-trained AraBERT model to convert Arabic sentences into semantic-rich embeddings. These vector representations are then passed to a quantum classifier to make predictions. Our research is structured into two main sections. The first investigates how different quantum data encoding strategies affect the performance of the QSVM. In the second section, we perform a comparative analysis of two hybrid quantum-classical learning approaches: the H-QNN and the VQC, focusing on their accuracy and efficiency. We benchmark all quantum models against classical classifiers to evaluate performance gains. The results highlight the potential of QML in Arabic NLP, offering promising directions for future research. By presenting this model, we aim to highlight how QML can be applied to Arabic language tasks, demonstrating its potential and paving the way for future explorations in this area.

2 Related work

Quantum machine learning is a growing area of study that's drawing increasing interest from the research community. Scientists are actively developing and experimenting with quantum-based algorithms to identify scenarios where they may provide advantages over classical approaches. One of the most computationally intensive tasks in this field is natural language processing, where the complexity and richness of human language pose significant challenges. Recent research has introduced various quantum architectures tailored for NLP tasks, including quantum convolutional neural networks [16], quantum Kernel methods [17], and quantum self-attention neural networks [18], all of which aim to harness the power of quantum computation to capture complex linguistic structures more efficiently. Additionally, hybrid classical-quantum transfer learning [19] has emerged as a promising approach, combining the strengths of classical models with quantum enhancements to improve generalization on limited-resource datasets. While much of the current focus has been on English-language processing, recent studies are beginning to extend these models to other languages, underscoring the potential of quantum models to support diverse linguistic and cultural contexts. In [20], the research compares the performance of a quantum support vector machine with a classical random forest method for detecting emotions in Arabic text. Two different datasets were used in the analysis, and both techniques achieved impressive scores in terms of precision, recall, and $F1$. Notably, quantum computing proved highly accurate when working with large datasets, while traditional methods delivered quicker results on smaller ones. These findings pave the way for developing more robust systems for Arabic document classification. The researchers in this study [21] address the challenges posed by the limited research interest in low-resource languages like Bengali,

primarily due to restricted computational resources. To address this issue, they implemented a quantum machine learning approach. The study introduces a quantum-classical hybrid architecture that integrates a pre-trained multilingual BERT model for word embeddings with two innovative quantum models, batch upload quantum recurrent neural network (BUQRNN) and parameter nonshared BUQRNN (PN-BUQRNN) to perform sentiment analysis on Bengali text. Experiments were conducted using two benchmark sentiment analysis datasets, and the results demonstrate that the proposed BUQRNN model delivers superior performance and computational efficiency compared to traditional QRNNs. Additionally, the findings reveal that the PN-BUQRNN model significantly outperforms both the BUQRNN and classical models in certain scenarios, highlighting its potential as a cutting-edge solution. Parallel efforts in recent research have extended quantum kernel methods to low-resource multilingual speech processing. For instance, [22] introduced a quantum multiple kernel learning (QMKL) framework for spoken command recognition, which combines multiple Gaussian quantum kernels to map acoustic features into enhanced quantum embeddings. Evaluated on multilingual speech data from several low-resource languages, their approach achieved an average accuracy gain of +5.28% over both classical RNNs and single-kernel quantum baselines, with the largest improvement observed under extreme data scarcity. These findings further support the idea that multi-kernel quantum architectures can effectively capture complex linguistic patterns in low-resource speech applications. To provide a clearer overview of existing efforts in quantum NLP for low-resource languages, Table 1 summarizes key studies, highlighting the quantum models employed, target tasks, datasets used, and reported outcomes.

3 Literature review

3.1 Quantum machine learning

Quantum machine learning (QML) is an emerging area that blends quantum computing with machine learning to tackle complex problems more efficiently. It focuses on developing quantum versions of classical learning algorithms and also uses machine learning methods to explore and interpret quantum systems. By making use of quantum properties like superposition and entanglement, to address problems that exceed the capacity of classical computation [23]. QML leverages the unusual principles of quantum mechanics, such as superposition and entanglement, to address challenges in traditional machine learning. By operating on many states at once, which makes the quantum computer can process and analyze data more efficiently than classical machines for certain tasks. This capability provides the potential for exponential speedups in solving machine learning problems. QML is all about developing algorithms that adapt standard machine learning methods for use in quantum environments. It covers both supervised learning, where models are trained on labeled datasets, and unsupervised learning, which focuses on detecting patterns in data without predefined labels. These techniques are frequently realized through quantum neural networks (QNNs) or quantum support vector machines (QSVMs) [24]. The machine

Table 1 Summary of recent quantum NLP studies focused on low-resource languages

Study	Language(s)	Quantum model	Task	Dataset	Results
Omar et al. (2023)	Arabic	Quantum SVM	Emotion detection	Twitter-based Arabic datasets	Outperforms classical models on large-scale datasets in terms of both accuracy and efficiency
Yang et al. (2023)	Multilingual	Quantum Multiple Kernel Learning (QMKL)	Spoken command recognition	Multilingual low-resource speech dataset	QMKL surpassed both classical and single-kernel quantum baselines, achieving an average accuracy improvement of 5.28%
WENBIN et al. (2024)	Bengali	Multilingual BERT + BUQRNN/ PN-BUQRNN	Sentiment analysis	Bengali sentiment analysis datasets	PN-BUQRNN outperformed QRNN and classical baselines in both accuracy and efficiency in some scenarios

The table highlights the quantum models used, their target languages and tasks, datasets, and key performance results

learning process is shown in Fig. 1, highlighting a comparison between classical and quantum approaches during the training and validation stages. A notable difference between QML and traditional machine learning is the inclusion of two extra steps: quantum encoding and measurement. Quantum encoding is essential for transforming classical data into quantum states, a necessary step in every QML process since the available data is typically classical, not quantum. Measurement is the process of obtaining classical information from a quantum system. Unlike classical systems, where measurements can be made without disturbing the system, quantum measurements fundamentally alter the state of the system. This is because quantum information is encoded in superpositions, which are delicate and collapse upon measurement. Quantum mechanics is inherently probabilistic, meaning that the outcomes of measurements are not deterministic but rather have associated probabilities. When performing a measurement on a single qubit in state $|\psi_i\rangle$, the likelihood of observing the result $|0\rangle$ is represented by $|\langle 0|\psi\rangle|^2$, while the chance of observing the result $|1\rangle$ is $|\langle 1|\psi\rangle|^2$ [25].

Quantum machine learning (QML) methods generally fall into three distinct categories, depending on how quantum concepts are applied within machine learning, the three main approaches are hybrid classical-quantum ML, quantum-inspired ML, and pure quantum ML [26].

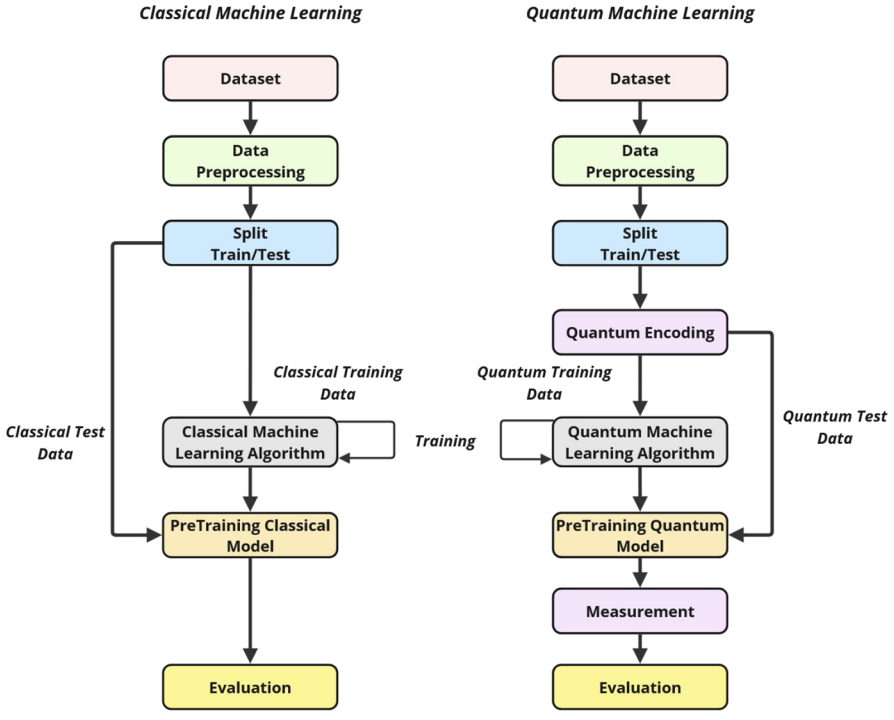


Fig. 1 Processing steps in classical and quantum machine learning during training and validation phases

Hybrid Quantum-Classical Machine Learning: It combines quantum computing with traditional machine learning techniques to enhance performance and address challenges that are difficult for classical machine learning alone. Although this approach is more widely used than other quantum machine learning methods, ongoing research aims to further improve its effectiveness. Meanwhile, several real-world applications have adopted this technique to boost the effectiveness of traditional machine learning, particularly in classification tasks [27, 28].

Quantum-Inspired Machine Learning (QiML): Quantum-inspired machine learning algorithms use ideas from quantum mechanics and computing while still running on standard classical machines. They are crafted to harness these quantum concepts to improve efficiency, handle more complex models, or provide better solutions [29]. These methods have been successfully applied to a range of machine learning problems, including classification tasks [30, 31].

Pure Quantum ML: Pure quantum machine learning algorithms perform all computations on quantum devices, in contrast to hybrid methods that integrate classical and quantum resources. This exclusive reliance allows them to tap into the special characteristics of quantum mechanics for processing tasks [32]. Although they promise significant improvements in speed and efficiency on paper, current quantum hardware limitations are a major barrier to their real-world implementation.

3.2 Quantum data encoding

Quantum encoding or quantum embedding is the process of mapping classical data into quantum states, which are mathematical entities in a Hilbert space representing quantum systems. This process is fundamental for leveraging quantum computing in tasks such as quantum machine learning, quantum optimization, and quantum cryptography. Quantum encoding utilizes the properties of quantum states, such as superposition and entanglement, to represent data in forms suitable for quantum computation [33, 34]. Common encoding techniques include basis encoding, amplitude encoding, and angle encoding.

3.2.1 Basis encoding

Basis encoding in quantum computing is a strategy where a classical binary string is directly assigned to a specific computational basis state in an n-qubit quantum system. For example, if you have a binary string $x = (b_1, b_2, \ldots, b_n)$, with each $b_i \in 0, 1$, this string corresponds to the quantum state $|x\rangle = |b_1, b_2, \ldots, b_n\rangle$. It is the simplest encoding scheme as it translates classical binary representations directly into quantum states [35]. For a classical binary dataset D consisting of M patterns $\{x_1, x_2, \ldots, x_M\}$, where each $x_m = (b_1, b_2, \ldots, b_N)$, $b_i \in \{0, 1\}$, the superposition of basis encoded states is expressed as:

$$|D\rangle = \frac{1}{\sqrt{M}} \sum_{m=1}^{M} |x_m\rangle \tag{1}$$

Here, every $|x_m\rangle$ signifies a quantum basis state linked to a particular binary input pattern.

3.2.2 Amplitude encoding

Amplitude encoding offers a strategy in quantum computing for converting classical data into quantum states by mapping each entry of a data vector to the corresponding amplitude in a quantum state. This approach is highly efficient because it allows one to encode a dataset of 2^n values with only n-qubit [36, 37]. For a normalized classical vector $X = \left[x_1, x_2, \ldots, x_{2n}\right]^T$ with $\sum_{i=1}^{2n} |x_i|^2 = 1$ where $x \in \mathbb{C}^2$, the mathematical representation is given as follows:

$$|\psi_x\rangle = \sum_{i=1}^{2n} x_i |i\rangle \tag{2}$$

Here, $|i\rangle$ stands for the basic states in an n-qubit system, and x_i are the components of the normalized classical vector.

3.2.3 Angle encoding

Angle encoding is a technique that embeds classical data into quantum systems by using phase shifts between quantum states. In this method, data points are represented as rotations on the complex plane. These rotations are carried out with quantum gates such as $R_x(\theta)$, $R_y(\theta)$, and $R_z(\theta)$, which act along the X, Y, and Z axes of the Bloch sphere, respectively. Here, the rotation angle θ is directly linked to a specific feature from the classical dataset, thus allowing the quantum circuit to process the information [38]. For angle encoding, if the classical information is represented by angles $\theta_1, \theta_2, \ldots, \theta_n$, the state of a quantum system is transformed when using the $R_x(\theta)$ rotation gate as:

$$|\psi\rangle = \bigotimes_{i=1}^{n} R_x(\theta_i) |0\rangle \tag{3}$$

3.3 Quantum support vector machine

Support vector machines (SVMs) are commonly applied in supervised learning, especially when tackling classification challenges [39]. Their main strategy involves finding a decision boundary known as a hyperplane that can clearly distinguish between different categories in the dataset. In cases where the data isn't easily separated in its original form, SVM maps it into a higher-dimensional space to discover a more effective dividing line [40]. To achieve this, SVM uses something called a kernel function to handle data in a more complex space without doing the transformation directly. This helps it separate the data better. Some common choices for these kernels are linear, polynomial, and RBF kernels. Given a dataset

$\mathcal{D} = \{(x_i, y_i) \mid x_i \in \mathbb{R}^d, y_i \in \{-1, 1\}, i \in \{0, 1, \ldots, n\}\}$, the decision function of an SVM using the kernel trick can be expressed as follows:

$$\hat{y}(x) = \text{sign}\left(\sum_{i=1}^{n} \alpha_i y_i K(x_i, x) + b\right) \tag{4}$$

where $\hat{y}(x)$ is the predicted label of the input data point x_i. The function returns either $+1$ or -1 , $K(x_i, x)$ is the kernel function. This construction can be achieved by maximize the following function:

$$\sum_{i=1}^{n} \alpha_i - \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} \alpha_i \alpha_j y_i y_j K(x_i, x_j) \tag{5}$$

where α_i is the Lagrange multipliers, $K(x_i, x)$ is the kernel function, and y Class label of the data point it can be $+1$ or -1.

In the quantum SVM we used quantum Kernel methods that map input data to high-dimensional vector spaces called reproducing kernel Hilbert spaces which transform the x_i to $|\psi_i\rangle$ quantum state. In these spaces, a linear function is learned and the dimension of the space can be infinite, making the method highly expressive. To manage the space of infinite dimensions, we apply the kernel trick. This technique efficiently computes the inner products of the high-dimensional vectors using the formula $K(x, x') = |\langle \phi(x) | \phi(x') \rangle|^2$, which in turn simplifies the learning process [41]. The QSVM method leverages a quantum encoding strategy paired with a SWAP test to compute its kernel, significantly reducing the qubit requirement and thus improving its scalability. This well-known quantum procedure assesses the similarity between two quantum states by determining their overlap. After this comparison, measurements are taken to capture the original states, with the SWAP gate playing a crucial role in evaluating the resemblance between them [42]. This SWAP test can be represented as follows:

$$\langle \text{SWAP} \rangle = \left| \langle \varphi(x_1) \otimes \varphi(x_2) | \text{SWAP} | \varphi(x_1) \otimes \varphi(x_2) \rangle \right|^2 \tag{6}$$

3.4 Variational quantum circuit

A variational quantum circuit, often known as an Ansatz, is a type of quantum setup that blends quantum processes with classical computing methods to work together effectively. It utilizes quantum circuits with adjustable parameters, implemented through tunable quantum gates, which are refined using classical optimization techniques. Variational quantum circuits are designed for use on noisy intermediate-scale quantum (NISQ) devices and are known for their flexibility, noise resilience, and capability to model complex functions [43, 44]. They are widely applied in quantum learning, optimization, and emulating quantum systems. Additionally, variational quantum circuits are widely used in various machine learning tasks, including classification [45, 46] and regression [47, 48].

3.4.1 IQP circuit

IQP circuits, or instantaneous quantum polynomial time circuits, are a common Ansatz in quantum computing. In these circuits, you start by setting up each qubit in either the $|+\rangle$ or $|0\rangle$ state. After this initialization, a Hadamard gate is applied to each qubit, effectively mixing the states. Next, the circuit implements a series of diagonal unitary operations $\mathcal{D}$, based on the input x_i. The entire process wraps up with a measurement in the Hadamard basis, capturing the final state of the system. To increase the depth of the circuit, the layer of diagonal unitaries can be repeated multiple times [49]. The Fig. 2 illustrates a graphical representation of the IQP circuit with two repetitions of the diagonal unitary operation.

In essence, the circuit applies Hadamard gates both before and after the diagonal unitaries, with the final step being a measurement in the Hadamard basis. For an n qubit system with two repetitions of the diagonal unitary operation, the overall transformation of the circuit can be expressed as follows:

$$|\Phi(x)\rangle = H^{\otimes n} D_{\Phi(x)} H^{\otimes n} D_{\Phi(x)} H^{\otimes n} |0\rangle^{\otimes n} \tag{7}$$

where $H^{\otimes n} i$ represents the Hadamard gate applied to n-qubit, x_i denotes the data input, and $\mathcal{D}$ refers to a diagonal unitary operation.

In machine learning, IQP circuits play a pivotal role in mapping features into high-dimensional quantum Hilbert spaces, enabling the use of kernel methods for classification tasks. By leveraging the intricate structure of quantum states, these circuits enhance the capabilities of data processing beyond the limits of classical approaches. This encoding facilitates the nonlinear transformation of classical data into quantum states. IQP circuits are believed to be classically hard to simulate, offering potential computational advantages in quantum machine learning. Notably,

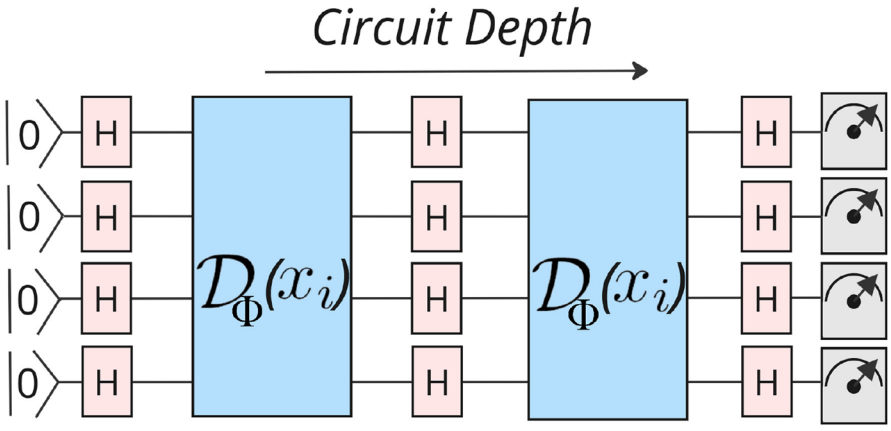


Fig. 2 A detailed graphical representation of an IQP circuit for a 4-qubit system, featuring two diagonal unitary layers

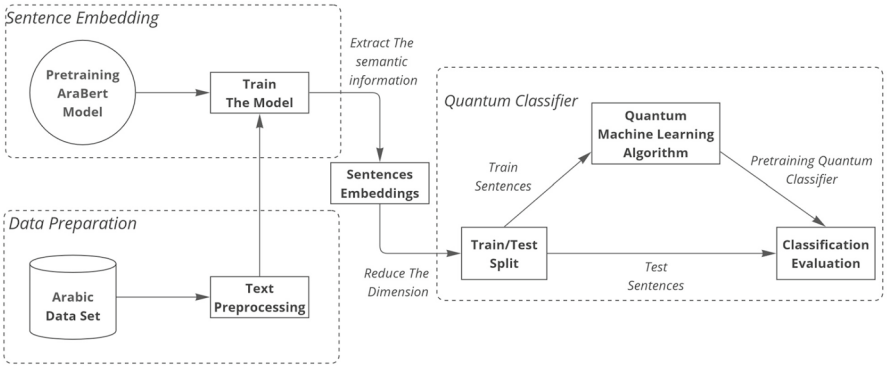


Fig. 3 AraBERT-QC architecture

they are instrumental in generating quantum kernels that are challenging to estimate using classical methods [50].

4 Methodologies and experiment

Our proposed architecture AraBERT-QC that integrates the AraBERT language model with a quantum classifier, aiming to address the complexities of the Arabic language through a novel quantum machine learning approach. Our architecture is shown in Fig. 3. We break our experimental work into two sections. In the first part, we investigate how various quantum data encoding methods affect both the accuracy and speed of the QSVM algorithm. In the second part, we compare the performance differences between training a variational quantum circuit directly and using a hybrid method that integrates parameterized quantum circuits with classical neural networks. All quantum components were implemented using the PennyLane framework[1] and tested on a PennyLane Lightning exact CPU-based simulator device with a 10-qubit size for the text classification task, enabling precise simulation of ideal quantum circuits.

4.1 Dataset

In this study, we utilized two distinct datasets to assess the performance of our integrated model. The first is the Hotel Arabic Reviews Dataset (HARD), introduced in [51], which comprises 93,700 reviews written in both Modern Standard Arabic (MSA) and various Arabic dialects, primarily for sentiment analysis tasks. The second dataset,[2] the Sarcasm Detection Dataset, contains Arabic YouTube comments annotated with binary labels indicating sarcastic (true) or non-sarcastic (false)

[1] https://pennylane.ai/.

[2] https://www.kaggle.com/datasets/farisalahmdi/arabic-youtube-comments-by-khalaya.

Table 2 The datasets used in the experiments

Dataset	All sentences	Class 1	Class 2
Sarcasm detection dataset	800	400	400
Sentiment analysis dataset	1000	500	500

Table 3 Sample Arabic sentences and their corresponding labels from the sarcasm detection and sentiment analysis datasets

Dataset	Text	Label
Sarcasm detection	المخيمات كانت في بداياتها بريئة مفيدة ثم تسيست	Sarcastic: Yes
	بعد سنوات سوف يقال بأن الكويت اسقطت برلين واسرت هتلر	Sarcastic: Yes
	انجازات الدولة العثمانيه انها نشرت الجهل في العالم الإسلامي وأسقطت الخلافة الاسلامية	Sarcastic: No
Sentiment analysis	انصح بالإقامة فيه. رائع وهادئ جدا انصح فيه والسعر ممتاز والافطار مميز	Sentiment: Positive
	الموقع والنظافه والخدمه المتميزه . اهتمام ممتاز بالنزلاء	Sentiment: Negative
	رائع. الشقة كانت واسعة ، كبيرة ، هادئة ونظيفة	Sentiment: Positive

content. Given the current limitations of quantum computing hardware, we selected balanced subsets of short texts from both datasets: 1,000 reviews from the sentiment analysis dataset and 800 comments from the sarcasm detection dataset. Both datasets were carefully cleaned to enhance the quality of the input. We stripped out punctuation and frequently used Arabic words that don't add value, which helped to reduce noise and stabilize how sentences are represented. After cleaning, we divide the data into training and testing groups using a 70% training and 30% testing split to allow a fair assessment of the model's effectiveness. The balanced distribution of samples in each dataset, as shown in Table 2, plays a crucial role in ensuring that the model is trained and evaluated without bias.

The Table 3 presents representative samples from the Sarcasm Detection and Sentiment Analysis datasets, highlighting the diversity of Arabic text and the labeling schemes used for training and evaluating the AraBERT-QC model.

Each sentence of the datasets was thoroughly cleaned by removing punctuation and Arabic stopwords. Each of those datasets was divided into a train set for training the quantum algorithms and a test set for evaluating the algorithms.

4.2 Sentences embeddings

Sentence embeddings, also called semantic vector space representations, which encapsulate the semantic meaning of sentences into vector representations, are crucial for natural language processing tasks. These embeddings capture various linguistic features, including syntactic structure, semantics, and the topic of the sentence [52]. In recent years, several language models have been developed that

can effectively represent sentences in semantic vector spaces. These models learn from large collections of text, allowing them to understand context effectively. One prominent architecture is the transformer-based framework, such as Sentence-BERT [53], which excels at generating high-quality sentence embeddings in many natural language processing tasks. In our study, we utilized AraBERT [54], a transformer-based model pre-trained specifically for the Arabic language, to generate high-quality sentence embeddings. AraBERT has been fine-tuned on a huge amount of Arabic text, which allows it to grasp the language's complex structure and its multiple dialects. We first applied tokenization using AraBERT's tokenizer. This step splits sentences into tokens. Each token is then converted into a high-dimensional embedding. To represent the entire sentence , we derived a fixed-length vector v_{sent} by aggregating the token embeddings, preserving both semantic and syntactic information. Each sentence from the datasets was processed through AraBERT to produce embeddings that effectively summarize the main meaning.

4.3 Dimension reduction

Embeddings generated by AraBERT are high-dimensional, creating significant computational challenges, especially when integrated with quantum machine learning models. Due to the constraints of existing quantum systems in the noisy intermediate-scale quantum (NISQ) era, which feature a small number of qubits prone to errors, managing dimensionality is crucial to ensure efficient processing and scalability. We applied principal component analysis (PCA) to reduce the dimensionality of the sentence embeddings and overcome this challenge. PCA is a method used to reduce the complexity of data by mapping it from a higher-dimensional space to one with fewer dimensions. It does so by finding key directions that represent the greatest variation in the data. Each sentence embedding, denoted as v_{sent}, was reduced to 10 principal components in our implementation, resulting in 10-dimensional vectors that retain the most significant information from the original high-dimensional AraBERT outputs. While this dimensionality reduction inevitably leads to some loss of information, this trade-off is necessary for adapting embeddings to the current limitations of quantum hardware and evaluating the feasibility of quantum models for text classification. This process simplifies the embeddings by keeping only the most relevant features, reducing both computation and the qubits needed for the quantum classifier. PCA is particularly suited to our task as it strikes a balance between dimensionality reduction and the retention of critical information, especially given the brevity of the sentences in our datasets. Reducing the complexity of the data helps quantum machine learning models run more effectively, improving their performance and making them more practical given the limitations of current quantum technology. This dimensionality reduction step is integral to the overall architecture, enabling the seamless integration of classical sentence embeddings into quantum models, thereby optimizing the hybrid quantum-classical approach for Arabic sentence classification. This method simplifies the computational workflow while fitting well with the limitations of the NISQ period, making it a viable option for building quantum-powered NLP models.

4.4 Quantum classifier

4.4.1 Quantum support vector classifier

The first quantum algorithm employed in our experiments is the QSVM. Rather than simply applying QSVM for text classification, our primary objective is to investigate whether different quantum encoding techniques can significantly influence its efficiency and performance. To this end, we used two distinct quantum encoding strategies to encode the sentence embeddings v_{sent}, which were previously extracted using AraBERT. These fixed-length vectors serve as compact representations of the input text, preserving both semantic and syntactic information. The first encoding approach is based on angle encoding, a straightforward yet effective feature map where each component of the sentence embedding $v_{\text{sent}} \in \mathbb{R}^n$ is mapped to a single-qubit Y-axis rotation. This transforms the initial state $|0\rangle^{\otimes n}$ into a quantum state that reflects the structure of the input data. The quantum state $|\psi\rangle$ is then represented as follows:

$$|\psi\rangle = \prod_{i=1}^{n} R_Y(v_{\text{sent}}[i])\,|0\rangle^{\otimes n} \tag{8}$$

The second encoding scheme is based on the Instantaneous Quantum Polynomial (IQP) embedding, which allows for more expressive representations. This method applies Hadamard gates before and after a data-dependent diagonal unitary operator, which captures pairwise interactions among the input features. The resulting quantum state is given by:

$$|\psi\rangle = H^{\otimes n} D(v_{\text{sent}}) H^{\otimes n} |0\rangle^{\otimes n} \tag{9}$$

where the diagonal unitary $D(v_{\text{sent}})$ is defined as:

$$D(v_{\text{sent}}) = \exp\left(i \sum_{i<j} v_{\text{sent}}[i]\, v_{\text{sent}}[j]\, Z_i Z_j \right) \tag{10}$$

Here, $Z_i Z_j$ denotes the tensor product of Pauli-Z operators acting on qubits i and j. Both encoding strategies were used to construct the quantum kernel of the QSVM. We selected them for their ability to encode sentence-level semantic information into quantum states while preserving structural dependencies. In contrast, we intentionally excluded amplitude encoding due to its complexity. Although it enables mapping into a quantum space of dimension 2^n, it requires zero-padding and normalization steps that can introduce computational overhead and potential information distortion. These limitations may hinder training efficiency and degrade model performance. To rigorously assess model robustness, to obtain the best parameters of the SVM, we employed five-fold cross-validation on a balanced dataset. The performance metrics, along with their standard deviations, were reported to provide a comprehensive evaluation.

4.4.2 Hybrid quantum-classical neural networks

Our architecture for developing a hybrid quantum-classical neural network model to classify Arabic text sentences is presented in Fig. 4.We trained this model on the training dataset using multiple batches. The steps of this architecture can be summarized as follows: (new)

1. **Classical Input Layer Initialization:** During the training phase, each input sentence is represented as a sentence embedding vector $\vec{V}$. These vectors are passed through a classical dense layer, which serves to reduce the dimensionality of the input and prepare it for quantum embedding. The weights in this layer are initialized using Xavier initialization to ensure stable gradients during backpropagation and efficient learning.
2. **Quantum Embedding and Encoding:** The reduced-dimension vector $\vec{h}$ output from the classical layer is then used to parameterize quantum rotation gates. Each component of $\vec{h}$ corresponds to one qubit, initialized in the $|0\rangle$ state. Information is encoded using angle encoding via R_y gates. This step transforms classical features into quantum states, resulting in:

$$|\psi_i\rangle = R_y(h_i)|0\rangle \tag{11}$$

   and collectively prepares the initial quantum state $|\psi\rangle$.
3. **Variational Quantum Circuit:** The quantum states are passed through a parameterized variational quantum circuit, denoted $C(\theta)$. The variational quantum circuit includes additional trainable R_y gates and entangling layers implemented via CNOT gates in a linear entanglement topology. The output quantum state is thus:

$$|\psi_{\text{out}}\rangle = C(\theta)|\psi\rangle \tag{12}$$

   This circuit is trained jointly with the classical layers, where the parameters θ are optimized during model training.

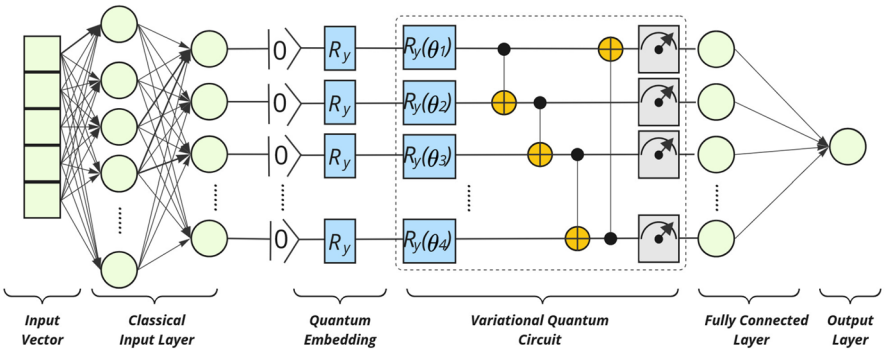


Fig. 4 The overall architecture of the proposed hybrid quantum-classical neural network model is designed as a quantum classifier for classifying Arabic sentences

4. **Measurement and Post-Processing:** Upon completion of the quantum operations, the qubits are measured in the Pauli-Z basis to extract the expectation values. These values are interpreted as classical features and passed through another dense (fully connected) classical layer.
5. **Output Layer and Training Strategy:** The dense layer concludes with a single output unit using a sigmoid activation function, suitable for binary classification. The entire hybrid architecture is trained using the AdamW optimizer and binary cross-entropy loss, with a learning rate of 0.001 and 100 training epochs.

4.4.3 Variational quantum classifier

This model relies entirely on a variational quantum circuit for classification, unlike hybrid approaches that combine quantum and classical neural networks. The goal is to assess how effective the variational quantum circuit is when used on its own, rather than integrating it with classical components. The Fig. 5 illustrates the encoding layer and the structure of our proposed variational quantum circuit model.

We begin by initializing n-qubit in the ground state $|0\rangle$. Applying Hadamard gates to each qubit places them into a uniform superposition. To encode classical data, we apply Y-axis rotations to each qubit, where the rotation angles are determined by the elements of the input vector $\vec{V}$. This step transforms each qubit into a quantum state $|\psi_i\rangle$ that reflects the corresponding classical feature. These encoded quantum states are then processed by a variational quantum circuit composed of a sequence of unitary operations. Among these are CNOT gates, which are used to create a circular entanglement structure among the qubits. Entanglement is introduced after encoding to establish correlations between qubits, enabling the circuit to capture interdependencies among features rather than treating each input dimension in isolation. This entanglement mechanism is essential for modeling complex relationships in data, allowing the quantum model to form richer and more structured internal representations. Such capacity is particularly beneficial in tasks like sentiment analysis and sentence classification, where semantic meaning often arises from the interaction between multiple features. Additionally, the R operation denotes a

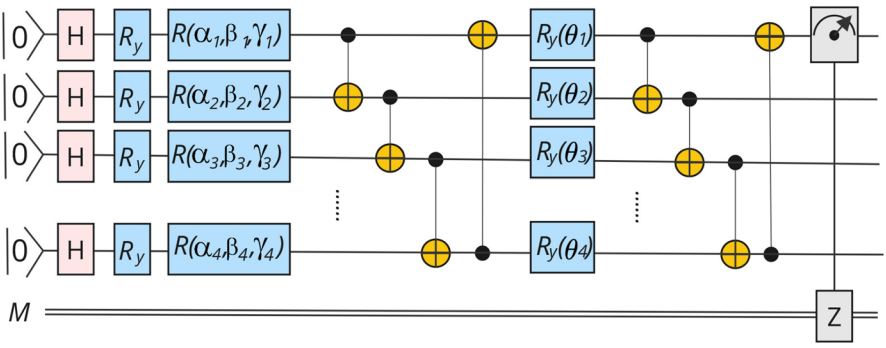


Fig. 5 The general architecture of the proposed variational quantum classifier is designed as a quantum classifier for classifying Arabic sentences

general single-qubit rotation, parameterized by three angles: α, θ, and γ. This gate allows flexible manipulation of a qubit's state through rotations around the Z and Y axes, enabling the construction of any desired single-qubit transformation. It can be represented as follows:

$$\mathrm{R}(\phi,\theta,\omega) = R_z(\omega)R_y(\theta)R_z(\phi) = \begin{bmatrix} e^{-i(\phi+\omega)/2}\cos(\theta/2) & -e^{-i(\phi-\omega)/2}\sin(\theta/2) \\ e^{-i(\phi-\omega)/2}\sin(\theta/2) & e^{i(\phi+\omega)/2}\cos(\theta/2) \end{bmatrix} \tag{13}$$

The VQC includes a parameterized operation around the Y-axis with the parameter θ to add more depth and complexity to the circuit.

Algorithm 1 demonstrates the training phase of the variational quantum circuit model. After transforming the Arabic sentences into numeric vectors $\vec{V}$ with d feature dimensions, these vectors are split into multiple batches and fed into the VQC model. We initialize random weights $\vec{\theta}$, both with fixed shapes. These weights are used within our proposed quantum circuit for training, while the biases are added at the circuit's output as part of the VQC model.

Algorithm 1 Training the quantum variational classifier

Input labeled training dataset $\mathcal{D} = \{(X_i, Y_i) \mid X_i \in \mathbb{R}^d, Y_i \in \{-1, 1\}, i \in \{0, 1, \ldots, n\}\}$, T: The number of iteration
Parameters: Initialize random weights $\vec{\theta}$
`backend` = QuantumSimulator
`Device` = **device**(`backend`, wires = d)
Split the data $\mathcal{D}$ into multiple $\vec{batches}$
Variational_Classifier: `Device`
Variational_Classifier($\vec{batch}$, $\vec{\theta}$):
    angle_encoding(input=$\vec{batch}$,wires=d)
    circuit($\vec{\theta}$)
    Return Expectation(**PauliZ**(wires=0))
for $i = 1$ to T **do**
  for $\vec{batch}$ **in** $\vec{batches}$ **do**
    $\hat{Y}$ = **Variational_Classifier**($\vec{batch}$,$\vec{\theta}$)
    Compute cost L using Y, $\hat{Y}$ and variational classifier model
    Optimizer Update model parameters $\vec{\theta}$ of variational classifier using gradient of L and learning rate η
  end for
end for
Return the trainable weights $\vec{\theta^*}$

The number of qubits in the circuit corresponds to the dimensionality of the feature vector, denoted as d. The variational quantum circuit outputs the expectation value of the Pauli-Z operator measured on a designated wire (e.g., wire 0). This value is used as the classifier's prediction, $\hat{Y}$. Model optimization is performed using a hinge loss function. The circuit is trained using a mini-batch strategy over 100 epochs, with a learning rate set to 0.005. The batch size is set to 100, and the

trainable parameters, $\vec{\theta}$, are updated iteratively using the Adam optimizer as implemented in the PennyLane framework. This training process continues until convergence. Upon completion, the optimized parameters $\vec{\theta}^*$, which minimize the cost function, are returned. These parameters define the trained variational quantum classifier. The runtime of each model was measured as the total wall-clock time from the start of the first epoch to the end of the final epoch. All models were trained under identical CPU-based conditions using the PennyLane Lightning simulator to ensure fairness and consistency.

5 Results and discussion

5.1 The effect of quantum encoding

The way classical data is mapped into quantum states plays a key role in the effectiveness of quantum machine learning models. This study explores the effects of two encoding methods, angle embedding and IQP embedding, on the QSVM for classifying short Arabic sentences. The quantum models in this study were evaluated using a quantum simulator running on a personal computer, ensuring a practical and realistic testing setup.

Table 4 presents a performance comparison of QSVM using different quantum encoding methods alongside a classical SVM with a radial basis function (RBF) kernel across two datasets: sarcasm detection and sentiment analysis. The results demonstrate the strong performance of the QSVM, achieving an accuracy of 86.25% on the sarcasm detection dataset and 83.67% on the sentiment analysis dataset. This

Table 4 Application's performance of AraBERT-QC with QSVM using angle and IQP embedding

Encoding	Dataset	Accuracy	Macro $F1$-score	Precision	Recall
Angle embedding	Sarcasm detection dataset	85.00 ± 0.0230	84.99 ± 0.0230	84.98 ± 0.0219	85.04 ± 0.0225
IQP embedding	Sarcasm detection dataset	**86.25 ± 0.0182**	**86.20 ± 0.0182**	**86.27 ± 0.0175**	**86.17 ± 0.0179**
Classical kernel	Sarcasm detection dataset	85.00 ± 0.0187	84.97 ± 0.0187	84.95 ± 0.0181	85.02 ± 0.0184
Angle embedding	Sentiment analysis dataset	**83.67 ± 0.0139**	**83.67 ± 0.0139**	**83.67 ± 0.0132**	**83.67 ± 0.0136**
IQP embedding	Sentiment analysis dataset	83.32 ± 0.0116	83.33 ± 0.0116	83.36 ± 0.0111	83.31 ± 0.0114
Classical kernel	Sentiment analysis dataset	83.00 ± 0.0107	83.00 ± 0.0107	83.01 ± 0.0103	82.98 ± 0.0105

Bold values indicate the best-performing results for each evaluation metric, highlighting the highest accuracy, $F1$-score, precision, or recall among the compared models

performance advantage comes from the ability of the quantum model to encode AraBERT-generated sentence embeddings into high-dimensional Hilbert spaces, preserving rich semantic and syntactic relationships. By mapping these embeddings to quantum states, the quantum kernel evaluates nonlinear similarities between states, capturing subtle linguistic patterns, such as contextual irony of sarcasm or sentiment polarity. These findings indicate that the quantum kernel employed in QSVM outperforms the classical RBF kernel SVM on both tasks. Among the encoding strategies evaluated, IQP embedding surpassed angle embedding in the sarcasm detection task, yielding higher accuracy and $F1$-scores. However, for the sentiment analysis, both embeddings delivered comparable performance. This suggests that the choice of quantum embedding significantly influences the effectiveness of the model in text classification and is highly dependent on the characteristics of the data set and the task context.

Selecting the right quantum encoding technique is essential for optimizing model performance according to dataset characteristics and application needs. For tasks requiring high precision and where computational resources are sufficient, IQP embedding offers substantial benefits. Conversely, for real-time applications or those requiring faster processing or operating under strict resource limitations, angle embedding can provide a practical balance. Overall, our findings highlight the importance of selecting appropriate quantum encoding techniques to maximize the potential of quantum-enhanced models in NLP tasks, particularly for low-resource languages such as Arabic.

The IQP embedding demonstrates significant inefficiency, particularly in terms of training time. As shown in Fig. 6, we present the duration of training of the QSVM model for both datasets. We exclude the classical baseline from this comparison, as the simulator introduces overhead and classical models run significantly faster, making such a comparison unfair given the current limitations of quantum hardware and

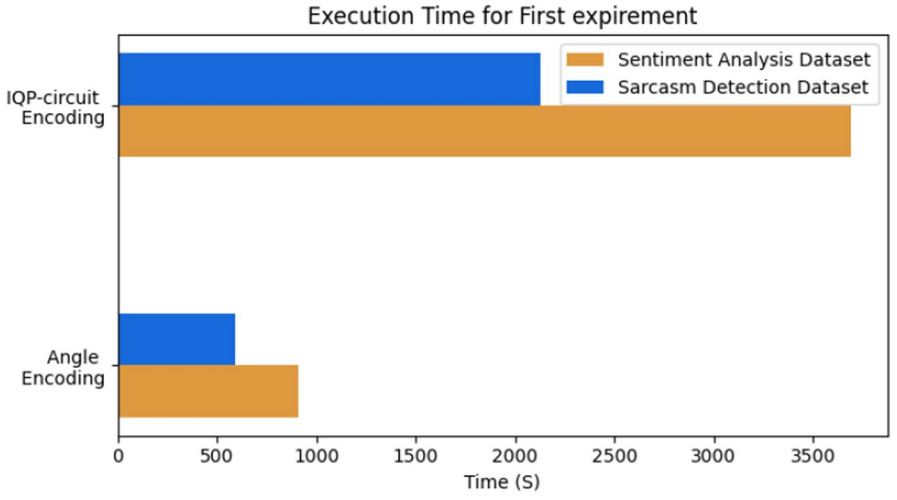


Fig. 6 Runtime comparison of QSVM with angle and IQP-circuit encoding techniques

simulators. Therefore, our evaluation focuses solely on comparing the performance of angle embedding and IQP embedding. The results clearly indicate that IQP embedding requires considerably more computational time due to its higher complexity. This trend is also evident in the sarcasm detection dataset. Training quantum models on classical simulators introduces a significant computational burden. This slowdown is expected due to the need to emulate quantum state evolution on classical hardware. It further reinforces our findings and highlights the importance of developing more efficient, quantum-native models as real quantum devices become more accessible. These observations underscore the importance of weighing computational costs against performance improvements when using more complex quantum encoding strategies like IQP embedding.

Notably, IQP embedding, while more complex and computationally intensive, demonstrated superior performance on the sarcasm detection task by extracting more nuanced features from the classical data compared to angle embedding. However, its advantages were less pronounced in the sentiment analysis task, where improvements were marginal. This highlights the trade-off between complexity and efficiency inherent in advanced quantum encoding methods. The computational cost of IQP embedding is particularly evident in its longer processing times and higher resource demands. Balancing accuracy and computational efficiency remains a key challenge in developing quantum machine learning models. In scenarios with limited computational resources or where faster processing is essential, such as in text classification tasks with high-dimensional inputs, angle embedding may be preferred. Although it has lower representational capacity, it provides a practical solution in resource-limited environments.

5.2 Comparison of VQC and H-QNN

This section presents a comparative analysis of three models: H-QNN, the VQC, and the classical neural network (Classical NN). The comparison focuses on their performance in sarcasm detection and sentiment analysis for Arabic text. The analysis highlights how the quantum-based models (H-QNN and VQC) utilize quantum principles in distinct ways, offering unique trade-offs in terms of computational efficiency and predictive accuracy when compared to their classical counterpart. Model performance was evaluated using accuracy, macro $F1$-score, precision, and recall across both datasets. The results, summarized in Table 5, illustrate the individual strengths of each model and underscore their complementary capabilities.

- **Sarcasm detection**: The VQC outperformed the H-QNN in terms of both accuracy and macro $F1$-score. The VQC achieved an accuracy of 85%, slightly surpassing the H-QNN's 84.17%. This indicates that the fully quantum design of VQC, using quantum entanglement and superposition, may be particularly effective in capturing the subtle linguistic patterns involved in sarcasm detection, where understanding intricate word-context relationships is crucial. However, the classical neural network yielded even better results on this dataset, attaining

Table 5 Comparative performance of VQC and H-QNN models on sarcasm detection and sentiment analysis datasets

Model	Dataset	Accuracy	Macro $F1$-score	Precision	Recall
H-QNN	Sarcasm detection dataset	84.17	84.13	84.13	84.13
VQC	Sarcasm detection dataset	85.00	84.98	84.96	85.00
Classical NN	Sarcasm detection dataset	**86.25**	**86.23**	**86.22**	**86.24**
H-QNN	Sentiment analysis dataset	83.00	83.00	83.02	82.98
VQC	Sentiment analysis dataset	**84.33**	**84.31**	**84.41**	**84.29**
Classical NN	Sentiment analysis dataset	82.33	82.30	82.41	82.29

Bold values indicate the best-performing results for each evaluation metric, highlighting the highest accuracy, $F1$-score, precision, or recall among the compared models

an accuracy of 86.25%, highlighting that quantum models do not consistently outperform classical approaches in all scenarios.

- **Sentiment analysis**: In the context of sentiment analysis, the VQC demonstrated superior performance, surpassing both the H-QNN and classical models, including a classical neural network and the QSVM. The VQC achieved an accuracy of 84.33%, while the H-QNN approach also outperformed traditional neural network with an accuracy of 83%. These results suggest that models leveraging quantum principles can achieve better performance, particularly on larger and more diverse datasets. This highlights the potential of hybrid quantum-classical models in addressing the complexities of the Arabic language, which is known for its rich morphology and diverse dialects.

The success of VQC can be attributed to its quantum-enhanced feature mapping, where AraBERT embeddings are projected into high-dimensional Hilbert spaces. This process preserves contextual word relationships and syntactic structures, enabling the variational quantum circuit to model complex decision boundaries through entanglement and interference. Classical models, limited by linear separability

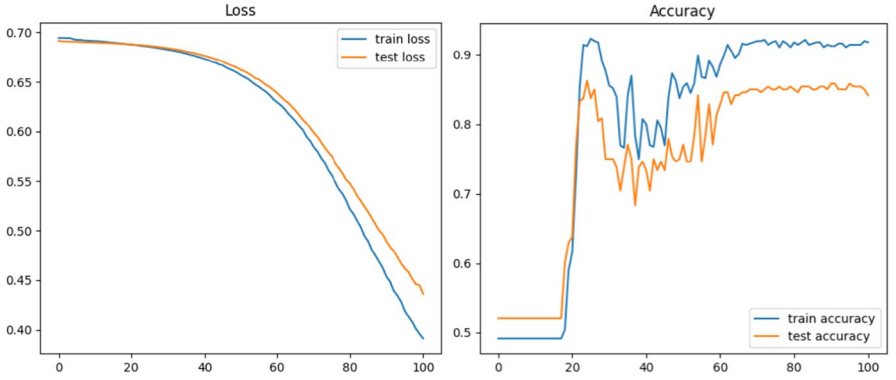


Fig. 7 Loss and accuracy progression across each iteration of the H-QNN on sarcasm detection dataset during training and testing

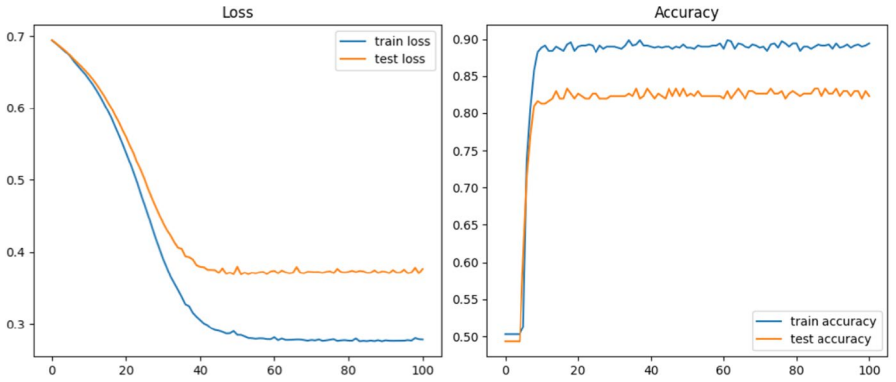


Fig. 8 Loss and accuracy progression across each iteration of the H-QNN on sentiment analysis dataset during training and test

constraints in the original feature space, struggle to replicate this capability. For sentiment analysis, where dialectal variations and morphological richness complicate classification. Figures 7 and 8 illustrate the progression of loss and accuracy for the H-QNN model across iterations for the sarcasm detection and sentiment analysis datasets, respectively, providing insight into the model's learning dynamics and generalization capabilities.

Similarly, Figs. 9 and 10 depict the cost progression and test accuracy during the training of the VQC model for both datasets, highlighting the performance differences between the two models.

The convergence behavior of the H-QNN and the VQC differs significantly across the sarcasm detection and sentiment analysis tasks, primarily due to architectural differences and the loss functions used. The H-QNN consistently demonstrates faster and more stable convergence than the VQC across both sarcasm detection and

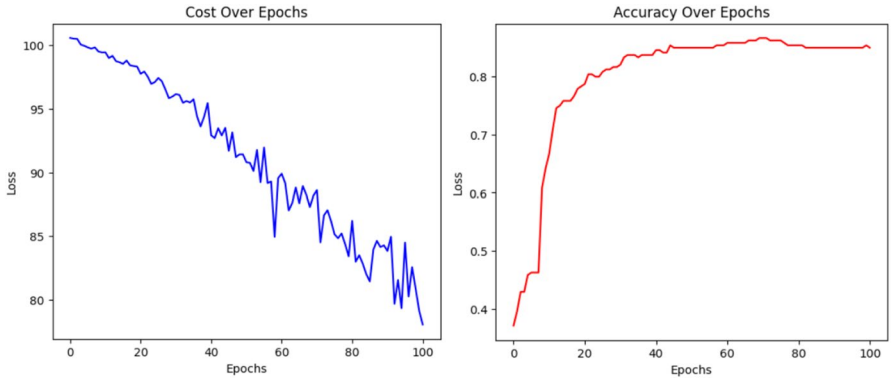


Fig. 9 Cost progression and test accuracy across each iteration of the VQC on sarcasm detection dataset during training

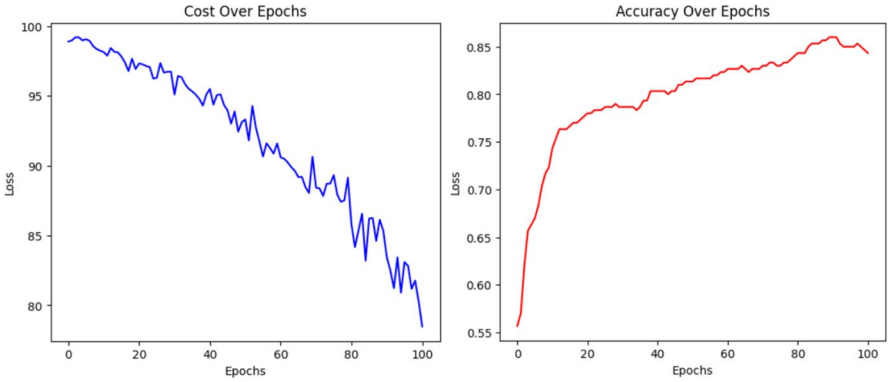


Fig. 10 Cost progression and test accuracy across each iteration of the VQC on sentiment analysis dataset during training

sentiment analysis tasks. In the sarcasm detection dataset, H-QNN reduces its loss from approximately 0.70 to 0.40 over 100 epochs, with significant improvements seen within the first 60 epochs, indicating effective learning and early convergence. Similarly, on the sentiment analysis dataset, the model converges rapidly, with loss dropping sharply from 0.70 to 0.30, mostly within the first 40 epochs. This reflects the capacity of the hybrid model to generalize well while refining its performance in later epochs. In contrast, the VQC exhibits a different convergence trend, influenced in part by its distinct loss function and purely quantum architecture. On both datasets, VQC starts with a slightly higher loss and reduces it more slowly. For sarcasm detection, the loss gradually declines and begins to plateau after approximately 60 epochs, signaling slower convergence. In the sentiment analysis task, while accuracy improves from 0.55 to 0.85 over 100 epochs, the loss curve flattens early, suggesting optimization difficulties. These include challenges such as vanishing gradients, barren plateaus, and limited circuit expressivity, common issues in variational quantum circuits.

The comparison between the VQC and the H-QNN, as illustrated in Fig. 11, demonstrates a notable difference in execution time across both datasets. This analysis omits the classical neural network due to the substantial computational overhead introduced by the quantum simulator, thereby emphasizing the performance trade-offs between quantum approaches. Despite the VQC potentially achieving higher accuracy on both the sentiment analysis and sarcasm detection tasks, it also exhibited greater computational efficiency. This improved efficiency can be attributed to the VQC's training strategy, which leverages a carefully designed variational circuit optimized through fewer but more impactful parameters. Specifically, the VQC uses 60 quantum gates, whereas the H-QNN employs only 20 quantum gates but relies on a large number of classical parameters. Moreover, the use of entanglement in VQC may have contributed to better feature representation with reduced circuit depth. In contrast, the H-QNN required significantly longer training times for both datasets. This is largely due to its architecture, which integrates quantum layers into a more extensive classical neural framework and was trained on the entire dataset. While

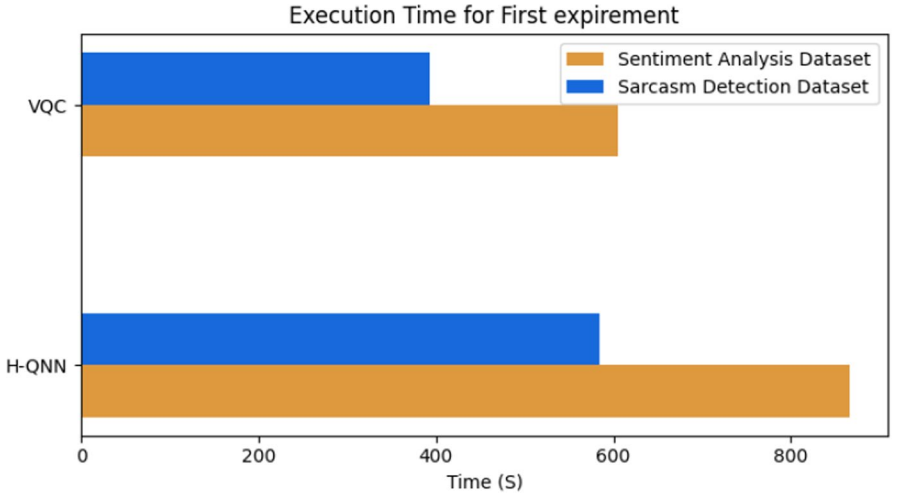


Fig. 11 Runtime comparison of VQC and H-QNN on the two sentiment analysis and sarcasm detection datasets

this may increase model capacity, it also introduces additional complexity and overhead during optimization. However, in terms of convergence behavior, the H-QNN demonstrated faster and more stable convergence than the VQC across both tasks. It is important to contextualize these findings within the current stage of quantum computing. While VQC is more efficient than H-QNN under these conditions, all classical models still train significantly faster than both quantum approaches. This is largely due to the immaturity of quantum hardware and the computational burden of quantum simulation. As quantum technologies continue to advance, this gap may narrow, but at present, classical machine learning remains more practical for large-scale or time-sensitive applications.

6 Conclusion

AraBERT-QC is a novel hybrid model that combines the strengths of classical deep learning and quantum computing to enhance the classification of short Arabic sentences. It integrates AraBERT, a transformer-based model pre-trained for Arabic, with quantum machine learning classifiers to address the challenges of processing morphologically rich and low-resource languages. Our experimental evaluation covered sentiment analysis and sarcasm detection using two balanced Arabic datasets. We examined the effect of quantum data encoding methods in QSVM, comparing angle and IQP embeddings. Both quantum variants outperformed the SVM with an RBF kernel, highlighting the strength of quantum kernels in capturing complex patterns. IQP embedding yielded the best results in sarcasm detection due to its expressive power, though at the cost of increased training time and circuit depth. In sentiment analysis, the performance gap was minimal, suggesting that embedding choice

should consider both task complexity and computational cost. In comparing hybrid quantum-classical models, the VQC demonstrated both computational efficiency and strong predictive performance. It achieved faster convergence and outperformed both the H-QNN and the classical neural network on the sentiment analysis task, highlighting its suitability for handling complex linguistic features in larger datasets. However, in the sarcasm detection task, the classical neural network achieved the best results, slightly surpassing both quantum models. These findings emphasize that while quantum-enhanced models do not universally outperform classical ones, they can provide tangible benefits in specific NLP scenarios. Such hybrid models could play an increasingly important role in processing low-resource languages like Arabic. An important contributing factor is the ability of quantum models to exploit superposition, entanglement, and interference. Superposition allows the parallel encoding and evaluation of multiple features, supporting the representation of ambiguous or context-dependent Arabic expressions. Entanglement captures complex dependencies between morphological components and enables parameter sharing across qubits, effectively reducing the number of trainable parameters needed in the circuit while maintaining expressive power. This compression produces more efficient models that are particularly well suited for deployment on near-term quantum devices. Interference refines decision boundaries during measurement, allowing for more nuanced and flexible classification outcomes, especially valuable in tasks like sarcasm detection, where semantic subtleties play a crucial role. Future research should focus on both advancing model architectures and addressing hardware challenges. More sophisticated quantum embeddings and variational circuits could further improve performance, but their scalability must be validated through evaluations on real quantum devices. Robust error mitigation techniques will be essential to handle decoherence, gate errors, and other limitations of NISQ hardware. At the same time, promising research directions include grammar-informed models such as DisCoCat, multilingual quantum-enhanced transformers, and novel Ansatz architectures tailored to morphologically rich languages like Arabic. Expanding these investigations to more diverse datasets and additional low-resource languages will also be critical for assessing the robustness and cross-lingual adaptability of QML approaches. A particularly promising direction is grammar-informed quantum models such as the DisCoCat framework, which translates grammatical structures into quantum circuits to produce more structured and semantically rich sentence representations. Although DisCoCat has been applied primarily to English, adapting this framework, or creating analogous models, for Arabic and other morphologically rich languages would represent a significant advancement in QNLP. Another promising direction is the development of multilingual quantum-based models, including quantum-enhanced transformers, capable of processing and classifying text across a wide range of languages. This would support the creation of more inclusive and globally applicable quantum NLP systems, especially for underrepresented and low-resource languages. Furthermore, designing novel Ansatz architectures specifically tailored to the syntactic and semantic patterns of morphologically complex languages like Arabic could enhance the expressive power of quantum models. Lastly, expanding these investigations to include more diverse datasets and additional low-resource languages is essential for evaluating the robustness and cross-cultural adaptability of

QML approaches. Closing the gap between simulation and real quantum hardware deployment will require collaborative efforts between linguists, quantum physicists, and computer scientists. As quantum technology matures, robust error mitigation techniques, hardware-efficient designs, and hybrid classical-quantum strategies will be key to enabling scalable and reliable quantum NLP systems.

Author contributions ID contributed to conceptualization, methodology, software, experimentation, and writing---original draft preparation and review & editing. HB helped in supervision, methodology, validation, and writing---review & editing.

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Data availability All data that support the findings of this study are included within the article.

Code availability All code will be made available upon reasonable request.

Declarations

Conflict of interest The authors declare no conflict of interest.

Ethical approval Not applicable.

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